

B.Tech. Degree III Semester Examination November 2018

IT/ME 1302 ELECTRICAL TECHNOLOGY
(2012 Scheme)

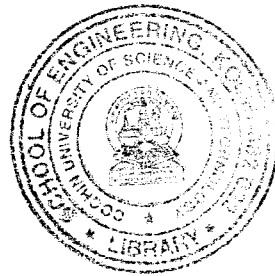
Time: 3 Hours

Maximum Marks: 100

PART A
(Answer *ALL* questions)

$(8 \times 5 = 40)$

- I.
 - (a) Draw and explain the vector diagram of a practical transformer on capacitive and inductive load.
 - (b) Discuss about various losses in a transformer and derive the condition for maximum efficiency.
 - (c) Derive the emf equation for alternator.
 - (d) What is commutation and what are the different methods to improve commutation in DC machines?
 - (e) Explain the effect of varying field excitation and load changes with the help of phasor diagram for synchronous motor.
 - (f) Explain why synchronous motors are not self starting.
 - (g) Explain the one line diagram of power system from generation to distribution.
 - (h) Explain about
 - (i) corona
 - (ii) skin effect
 - (iii) ferranti effect
 - (iv) types of AC transmission



PART B

$$(4 \times 15 = 60)$$

- II. (a) The following results were obtained on a 50 KVA, 2400/120 V transformer. (12)
- (i) OC Test (LV side)
396 watts, 9.65 A, 120 V
- (ii) SC Test (HV side)
810 watts, 20.8 A, 92 V
- Calculate: (i) the circuit constants (ii) efficiency at full load 0.8 pf lagging (iii) voltage regulation.
- (b) Explain with diagrams open circuit test and short circuit test on transformer. (3)

OR

- III. (a) Explain the working of auto transformer and also derive an expression for saving of copper when compared to ordinary transformer. (5)
- (b) A 100 kVA, 50 Hz, 440 V/11000 V single phase transformer has an efficiency of 98.5% when supplying full load current at 0.8 power factor lagging and an efficiency of 99% when supplying half full load current at unity power factor. Find the core losses and copper losses corresponding to full load current. At what value of load current will the maximum efficiency be attained? (10)

(P.T.O.)

- IV. (a) Explain the various speed control methods of DC shunt motor. (5)
 (b) A 200 V DC series motor runs at 1000 rpm and takes 20 A. Combined resistance of armature and field is 0.4Ω . Calculate the resistance to be inserted in series so as to reduce the speed to 800 rpm, assuming torque to vary as square of the speed and linear magnetization curve. (10)

OR

- V. (a) Explain the power flow diagram of DC generators. (5)
 (b) Explain the different characteristic curves of DC generators. (5)
 (c) An 8 pole DC generator has 500 armature conductors and has a useful flux of 0.065 wb. What will be the emf generated if it is lap connected and runs at 1000 rpm? What must be the speed at which it is to be driven to produce same emf if it is wave wound? (5)

- VI. (a) Explain the distribution factor and pitch factor with respect to alternators. (5)
 (b) Explain about emf method for voltage regulations. (3)
 (c) A 3 phase star connected alternator is rated at 1600 KVA, 13.5 KV. The armature resistance and synchronous reactance are 1.5Ω and 30Ω respectively per phase. Calculate the percentage regulation for a load of 1280 kW at 0.8 loading power factor. (7)

OR

- VII. (a) Derive the torque equation for a 3 phase induction motor and also find the expression for maximum starting torque. (5)
 (b) A 3 phase, 500 V, 50 Hz induction motor with 6 poles gives an output of 20 kW at 950 rpm with a pf of 0.8. The mechanical losses are equal to 1 kW. Calculate for this load (i) slip (ii) rotor copper loss (iii) input, if the stator losses are 1500 W (iv) line current. (10)

- VIII. (a) Explain thermal power plant with diagram. (12)
 (b) Discuss the merits and demerits of nuclear power plant. (3)

OR

- IX. (a) Discuss in detail: (7)
 (i) Isolators
 (ii) Circuit breakers
 (iii) Bus bar
 (iv) Potential transformer
 (v) Current transformer
 (b) Discuss about different types of substations. (5)
 (c) Discuss about various DC transmission schemes. (3)

